

Web Development

HTML Fundamentals + CSS Styling • Day 1 – 3

What We'll Cover

01

HTML Foundations

DOCTYPE, structure, headings, paragraphs

02

Text & Lists

Formatting tags, ordered/unordered/description lists

03

Links, Images, Tables

Hyperlinks, , <table>

04

Forms & Media

Inputs, form elements, video/audio

05

Semantic HTML

<header>, <footer>, <nav>, <section>, <article>

06

Layout Basics

div, span, block vs inline, attributes

07

CSS Intro

Syntax, selectors, application methods

08

CSS Properties

Typography, backgrounds, box model, positioning

01

HTML Foundations

What is HTML & how a document is structured

What is HTML?

Role of HTML

- Defines the structure and layout of web pages
- Uses elements (tags) to mark and describe content
- Tells browsers how to display content (headings, links, images...)
- Makes content accessible and searchable (SEO)

```
<!-- The simplest HTML page -->
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Hello, World!</h1>
    <p>This is my first page.</p>
  </body>
</html>
```

HTML = HyperText Markup Language

- HyperText → links between pages
- Markup → annotates content with tags like <h1>, <p>, <a>
- Language → structured rules browsers understand

... Browser renders →

Hello, World!

This is my first page.

HTML Document Structure

`<!DOCTYPE html>`

Declares this is HTML5 to the browser

`<html>`

Root element — wraps the entire page

`<head>`

Metadata: title, charset, stylesheet links (not visible to users)

`<body>`

All visible content: headings, text, images, buttons...

Headings <h1> to <h6>

headings.html

```
<h1>Main Page Title</h1>
<h2>Section Heading</h2>
<h3>Sub-section</h3>
<h4>Sub-sub heading</h4>
<h5>Minor heading</h5>
<h6>Smallest heading</h6>
```

● ● ● Browser Output

Main Page Title

Section Heading

Sub-section

Sub-sub heading

Minor heading

Smallest heading

Key Points

- h1 = most important (one per page, for SEO)
- h2–h3 used for sections & sub-sections
- h4–h6 rarely needed; don't skip levels
- Headings help SEO & screen-reader navigation

Paragraphs, Line Breaks & Horizontal Rules

`<p>`

Paragraph — adds space before & after content

Wraps a block of text into a standalone paragraph

`<br>`

Line break — moves content to next line

Self-closing tag; no new paragraph created

`<hr>`

Horizontal rule — draws a dividing line

Self-closing; used to separate sections visually

example.html

```
<p>This is the first paragraph. It has space around  
it.</p>  
<p>Second paragraph starts here.<br>This line uses a  
break tag.<br>Another line.</p>  
<hr>  
<p>Content after the divider line.</p>
```

● ● ● Browser Output

This is the first paragraph. It has space around it.

Second paragraph starts here.

This line uses a break tag.

Another line.

Content after the divider line.

Text Formatting Tags

`<b>`

Bold (visual only)

Sample Text

`<strong>`

Bold + semantic importance

semantic

Sample Text

`<i>`

Italic (visual only)

Sample Text

`<em>`

Italic + semantic emphasis

Sample Text

`<u>`

Underline (visual only)

Sample Text

← Tag Description

Preview →

02

Lists in HTML

Ordered, Unordered & Description Lists

HTML Lists — , , <dl>

 Ordered

Items are numbered
Use when sequence matters
E.g. Steps, Rankings

```
<ol>
  <li>Boil water</li>
  <li>Add pasta</li>
  <li>Drain</li>
</ol>
```

● ● ● Ordered list output

1. Boil water
2. Add pasta
3. Drain

 Unordered

Items are bulleted
Use when order doesn't matter
E.g. Features, Grocery list

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

● ● ● Unordered list output

- HTML
- CSS
- JavaScript

<dl> Description

Term + description pairs
Use for glossaries, FAQs
<dt> term / <dd> definition

```
<dl>
  <dt>HTML</dt>
  <dd>Markup language</dd>
  <dt>CSS</dt>
  <dd>Style language</dd>
</dl>
```

● ● ● Description list output

HTML
Markup language
CSS
Style language

03

Links, Images & Tables

Connecting pages • Embedding media • Displaying data

Hyperlinks <a>

Internal Link

```
<a href="about.html">About Us</a>
```

Navigates to another page on the same site

External Link

```
<a href="https://google.com">Google</a>
```

Navigates to a different website

New Tab

```
<a href="..." target="_blank">Open</a>
```

Opens the link in a new browser tab

Email Link

```
<a href="mailto:hi@email.com">Email</a>
```

Opens user's email app with recipient pre-filled

Images `<img src="" alt="">`

Key Attributes

- `src` → file path or URL of the image
- `alt` → descriptive text if image fails to load (accessibility + SEO)
- `<img>` is self-closing — no closing tag needed
- `width` / `height` control display size (or use CSS)

```
<!-- Local image -->


<!-- Web URL -->


<!-- Missing image shows alt text -->

```

Attribute Breakdown

<code>src</code>	<code>=</code>	<code>"images/dog.jpg"</code>	Path to image file
<code>alt</code>	<code>=</code>	<code>"A cute dog"</code>	Accessibility text
<code>width</code>	<code>=</code>	<code>"300"</code>	Display width in px
<code>height</code>	<code>=</code>	<code>"200"</code>	Display height in px

• • • If image loads

🖼️ [image displays here]

• • • If image fails

Product photo

(alt text shown instead)

Tables <table>

<table>

Defines the whole table

<tr>

Table Row

<th>

Table Header cell (bold, centered)

<td>

Table Data cell

colspan

Cell spans multiple columns

rowspan

Cell spans multiple rows

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td>Alice</td>
    <td>25</td>
    <td>Mumbai</td>
  </tr>
  <tr>
    <td>Bob</td>
    <td>30</td>
    <td>Delhi</td>
  </tr>
</table>
```

● ● ● Browser Output

Name	Age	City
Alice	25	Mumbai
Bob	30	Delhi

04

Forms & Media

Input types, form elements, attributes, video & audio

Input Types

`type=
"text"`

Plain text entry

```
<input type="text" placeholder="Enter  
name">
```

Single selection

`type=
"radio"`

```
<input type="radio" name="g"  
value="f"> Female
```

`type=
"email"`

Email addresses

```
<input type="email"  
placeholder="you@mail.com">
```

File upload

`type=
"file"`

```
<input type="file">
```

`type=
"password"`

Hidden entry

```
<input type="password"  
placeholder="Enter password">
```

Submit the form

`type=
"submit"`

```
<input type="submit" value="Register">
```

`type=
"checkbox"`

Multiple selections

```
<input type="checkbox"> Option A
```


Form Elements & Attributes

Form Elements

- `<form>` — wraps all form controls; has action & method
- `<label>` — labels an input (for= pairs with id=)
- `<select>` — dropdown list; uses `<option>` inside
- `<textarea>` — multi-line text input (rows= cols=)
- `<button>` — clickable button (type="submit")

Form Attributes

- `required` — makes field mandatory
- `placeholder` — hint text inside input box
- `name` — key sent to the server (backend)
- `value` — default / initial value
- `method` — get or post (HTTP method)
- `action` — URL/API endpoint where data is sent

```
<form action="/register" method="post">
  <label for="uname">Username:</label>
  <input type="text" id="uname" name="username"
    placeholder="Enter username" required>

  <label for="country">Country:</label>
  <select id="country" name="country">
    <option>India</option>
    <option>USA</option>
  </select>

  <textarea name="bio" rows="3" placeholder="About you"></textarea>
  <button type="submit">Register</button>
</form>
```

Media Tags — <video> & <audio>

<video> Embed Video

```
<video src="clip.mp4"
  controls
  loop
  muted
  width="400">
</video>
```

controls → show play/pause bar
loop → replay automatically
muted → start without sound

● ● ● Video browser output

▶ 00:00 ————— 🔊]

VIDEO PLAYER |

<audio> Embed Audio

```
<audio src="music.mp3"
  controls>
</audio>

<!-- Multiple formats -->
<audio controls>
  <source src="s.mp3">
  <source src="s.ogg">
</audio>
```

● ● ● Audio browser output

▶ 00:00 ————— 🔊]

(audio player bar)

05

Semantic HTML

`<header>` `<footer>` `<nav>` `<section>` `<article>` `<aside>`

Semantic Tags — Why They Matter

`<header>`

Introductory content, site logo, main heading

`<article>`

Self-contained content: blog post, news item

`<nav>`

Block of navigation links (menus, pagination)

`<aside>`

Tangentially related: sidebar, ads, related links

`<section>`

Standalone thematic group of content (has a heading)

`<footer>`

Footer info: copyright, author, contact links

Semantic Layout — Visual Structure

Page

<header> Logo + Nav

<nav> Home | About | Contact

```
<main>
  <article>
    Blog Post
  </article>
```

```
<aside>
  Links,
  Ads
```

<section> Related Articles

<footer> © 2025 My Site

```
<!DOCTYPE html>
<html>
<head>...</head>
<body>

  <header>
    <nav>
      <a href="/">Home</a>
      <a href="/about">About</a>
    </nav>
  </header>

  <main>
    <article>
      <h2>Blog Post Title</h2>
      <p>Post content here...</p>
    </article>
    <section>
      <h3>Related Posts</h3>
    </section>
  </main>

  <aside>
    <p>Advertisement</p>
  </aside>

  <footer>
    <p>© 2025 My Site</p>
  </footer>

</body>
</html>
```

06

Layout Basics & Common Attributes

div, span, id, class, style, block vs inline

div vs span & Block vs Inline

<div> Block Container

- Block-level: starts on a new line
- Takes full available width
- Used for page sections & layout
- Styled with CSS for layout

 Inline Container

- Inline: stays within the text flow
- Takes only as much width as needed
- Used to style part of text or group
- Cannot contain block elements

```
<div style="background: lightblue; padding: 10px;">
  This is a block div — full width
  <span style="color: red; font-weight: bold;">red
  bold</span>
  and this continues inline.
</div>
<div style="background: lightyellow; padding: 10px;">
  Second div starts on a new line.
</div>
```

● ● ● Output

This is a block div — full width
red bold and this continues inline.

Second div starts on a new line.

Property	Block	Inline
New line?	Yes	No
Width	Full width	Content only
Examples	div, p, h1–h6	span, a, em

Common Attributes — id, class, style, src, href, alt

id

Unique identifier for an element

```
<h1 id="main-title">Title</h1>
```

Used in CSS (#main-title) and JS

class

Groups elements for shared styling

```
<p class="highlight">Text</p>
```

Used in CSS (.highlight) — reusable

style

Inline CSS styling

```
<p style="color:blue;">Blue text</p>
```

Quick styling; external CSS preferred

src

Source file for media/scripts

```

```

Used in , <script>, <iframe>

href

URL for hyperlinks/stylesheets

```
<a href="https://site.com">Click</a>
```

Used in <a> and <link> tags

alt

Alternative text for images

```

```

Accessibility + SEO — always add this!

Comments <!-- --> & Summary

HTML Comments <!-- -->

```
<!-- This is a comment — not visible to users -->  
<h1>Page Title</h1> <!-- Main heading -->  
<!-- TODO: Add contact form here -->
```

Key Takeaways



HTML = structure of web pages



Forms collect user input (action, method)



Semantic tags improve accessibility & SEO



id = unique | class = reusable groups



Tags organize content semantically



Media tags embed video & audio



block = full width | inline = content width



alt text = accessibility for images

07

CSS — Styling HTML

Syntax, selectors & how to apply CSS

What is CSS?

CSS = Cascading Style Sheets

- HTML = structure | CSS = appearance
- Applies colors, fonts, layout, spacing, animations
- Separates design from content for easier maintenance
- One CSS file can style hundreds of HTML pages

3 Ways to Apply CSS

- Inline → `style=""` attribute directly on element
- Internal → `<style>` tag inside `<head>`
- External → separate `.css` file (BEST PRACTICE)

example.html

```
<!-- 1. Inline -->
<p style="color: red; font-size: 18px;">Red
text</p>

<!-- 2. Internal -->
<head>
  <style>
    p { color: blue; font-size: 16px; }
  </style>
</head>
```

style.css

```
/* style.css */
p {
  color: blue;
  font-size: 16px;
  line-height: 1.6;
}

h1 {
  color: navy;
  text-align: center;
}
```

CSS Syntax & Selectors

CSS Syntax

```
p { color : red ; }
```

Selector

Property

Value

Element

```
p { }
```

Targets all <p> tags

div, h1, a, img, ...

Class

```
.menu { }
```

Targets elements with class=menu

Multiple elements, reusable

ID

```
#header { }
```

Targets element with id=header

One element only, most specific

08

CSS Properties

Typography • Backgrounds • Box Model • Positioning • Pseudo-classes

CSS Typography Properties

font-family

Sets font style

```
font-family: "Arial", sans-serif;
```

font-size

Size of text

```
font-size: 18px;
```

color

Text color

```
color: #2563EB;
```

font-weight

Thickness

```
font-weight: bold; /* or 400, 700 */
```

text-align

Alignment

```
text-align: center; /* left|right|justify */
```

text-decoration

Underline / strikethrough

```
text-decoration: underline;
```

line-height

Vertical spacing between lines

```
line-height: 1.6;
```

```
p { font-family: Arial; font-size: 16px; color: navy; font-weight: bold; text-align: center; line-height: 1.2; }
```

Bold centered navy text

CSS Background Properties

background-color

Sets background color

```
background-color: #DBEAFE;
```

background-image

Applies image as background

```
background-image: url('bg.jpg');
```

background-size

Scales background image

```
background-size: cover; /* or contain */
```

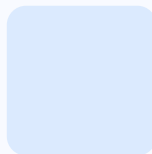
background-repeat

Controls image repeat

```
background-repeat: no-repeat;
```

style.css

```
div {  
  background-color: #1E40AF;  
  background-image: url('pattern.png');  
  background-size: cover;
```



#DBEAFE



#93C5FD



#3B82F6

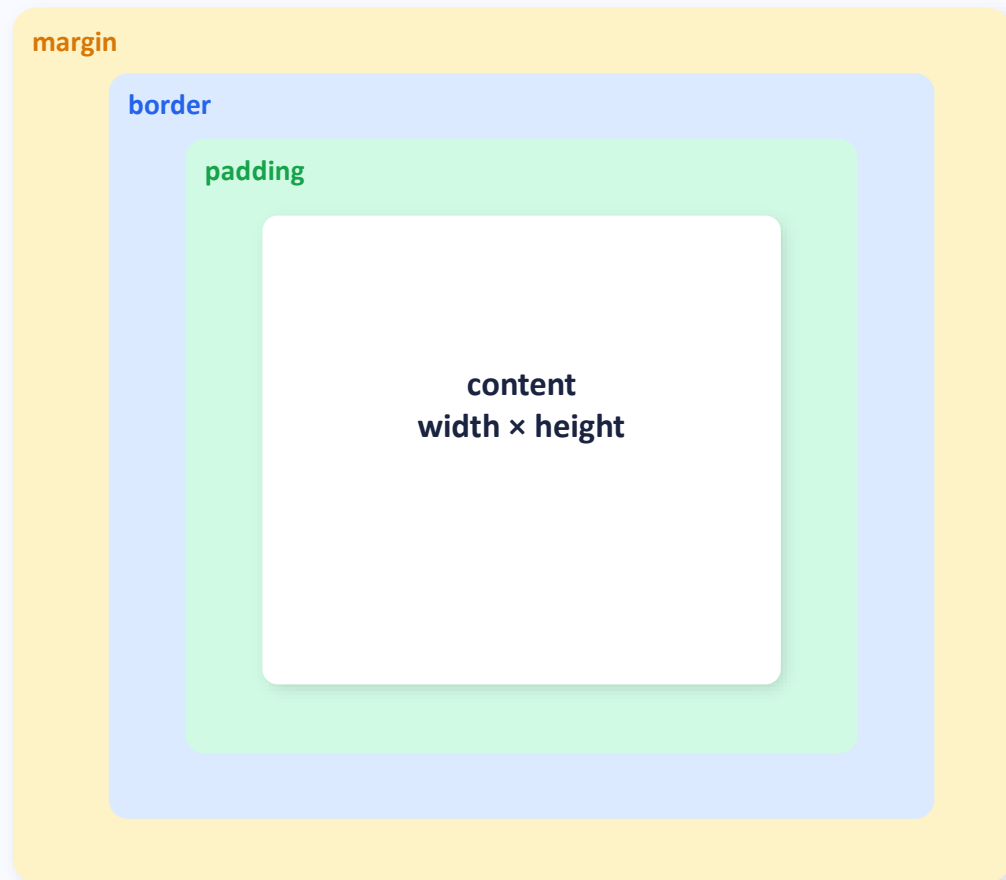


#1D4ED8



#1E3A8A

The CSS Box Model



margin

Space outside the border (transparent)

border

Outline around the element

padding

Space between content & border

content

The actual text/image area (width × height)

`style.css`

```
.box {  
  width: 200px;  
  height: 100px;  
  padding: 20px;  
  border: 3px solid navy;  
}
```


Pseudo-classes, Display & Positioning

Pseudo-classes

```
a:hover { color: red; }
a:visited { color: purple; }
}
button:active {
  background: darkblue;
}
```

Display Property

```
.block { display: block; }
.inline { display: inline; }
.flex-container {
  display: flex;
  gap: 10px;
}
```

Positioning

```
.a { position: static; }
.b { position: relative;
  top: 10px; }
.c { position: absolute;
  left: 0; top: 0; }
```

float & clear

float: left / right → moves element so text wraps around it | clear: both → stops wrapping

Position value	Behavior
static	Default — follows normal flow
relative	Offset from its normal position
absolute	Positioned relative to nearest positioned ancestor
fixed	Fixed to the viewport (doesn't scroll)

You've covered

HTML + CSS Foundations

✓ HTML structure & document anatomy

✓ Headings, paragraphs, formatting tags

✓ Lists, links, images, tables

✓ Forms, inputs, media tags

✓ Semantic HTML5 tags

✓ div, span, block vs inline

✓ CSS syntax, selectors & application

✓ Typography, backgrounds, box model

✓ Pseudo-classes, display & positioning